

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Summer 2024

Course: B. Tech Branch: Electrical Engineering and Allied Semester: IV

Subject Code & Name: Analog and Digital Electronics BTBS404

Max Marks: 60

Date: 20/06/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Solve Any Two of the following.

- A) Explain the various operations of regions of a transistor in CE configuration on its output characteristics. **06**
- B) What are different methods of transistor biasing? Explain any one in detail. **06**
- C) The h parameters of a transistor used in a single stage amplifier circuit are $h_{ic} = 1100$, $h_{rc} = 1$, $h_{fe} = -51$, $h_{oc} = 25 \mu A$. Determine the amplifier parameters for CC configuration when $R_s = R_L = 10 k\Omega$. **06**

Q.2 Solve Any Two of the following.

- A) What are various characteristics of an ideal Op-Amp? Discuss in details. **06**
- B) Draw the block diagram of an Op-amp and explain the purpose of using each block. **06**
- C) Explain Op-Amp as a differentiator with neat circuit diagram and necessary equations. **06**

Q. 3 Solve Any Two of the following.

- A) Convert the following numbers: **06**
- a) $(1101101.101)_2 = (?)_{10}$
 - b) $(126.75)_{10} = (?)_8$
 - c) $(375.75)_{10} = (?)_{16}$
- B) Solve the following arithmetic operations **06**
- a) Subtract $(15)_{10}$ from $(10)_{10}$ using 2's complement method of binary subtraction.
 - b) Subtract $(14)_{10}$ from $(18)_{10}$ using 1's complement method of binary subtraction.
- C) Explain all logic gates with their symbols, output expression and truth table. **06**

Q.4 Solve Any Two of the following.

- A) Explain TTL NAND gate circuit (Totem Pole Arrangement) with the help of circuit diagram. **06**
- B) What is meant by MOS logic family? Explain working of NAND gate by NMOS logic. **06**
- C) Explain J-K flip-flop with circuit diagram and truth table. **06**

Q. 5 Solve Any Two of the following.

- A) Reduce the expression for $f(A, B, C, D) = \sum_m (0, 1, 2, 3, 5, 7, 8, 9, 11, 14)$ using K-map. **06**
- B) Explain half adder circuit with the truth table. **06**
- C) Reduce the expression for $f(A, B, C, D) = \pi M (2, 3, 4, 5, 6, 7, 8, 11, 12)$ **06**

***** End *****